Umair Javaid Manj

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Research Experience

Doctoral Researcher	**Auckland University of Technology**, Health & Artificial Intelligence, Auckland • Building an AI based biomarker for chronic low back pain from cortical responses. • Translated the Zurich paraspinal vibration fMRI paradigm (Cole et al., 2022; Dörig et al., 2026) to 64 channel EEG: 80 Hz vibration at nine levels from T3 to S1, painful vs healthy segment in the same person. • Designed a case control study (34 CLBP, 34 pain free), sized in G*Power; planned sLORETA source estimation, representational similarity analysis and mixed models. • Ran Pilot 1 (one participant, 75 placements); traced a 92 to 99% decoding result to unequal analysis windows and withdrew it. • Produced a two minute protocol animation (on the research page above).	Oct 2025 to Present
Data & AI Governance Analyst	**Centre for Chiropractic Research, New Zealand College of Chiropractic**, Auckland • Collect and analyse data for the Centre's chiropractic research studies. • Manage participant data and consent records under the Privacy Act 2020 and ethics approvals.	Aug 2026 to Present
Program Manager	**Coding, Robotics & AI Learning Project**, NUST, Islamabad • Ran a national training program in AI, networking and cybersecurity, coordinating 80 instructors.	Sept 2024 to Oct 2025
Instructor	**HISDP, AI Lounge & Murabbi**, NUST, Islamabad • Supervised 18 student term projects; awarded Best Instructor.	May 2023 to July 2024
Research Associate	**COGNET Lab, SEECS**, NUST, Islamabad • Hardened deep learning models against adversarial attacks; mentored junior researchers.	Dec 2022 to Apr 2023
Research Assistant	**COGNET Lab, SEECS**, NUST, Islamabad • Built an end to end auditable system combining cryptography and computer vision.	Aug 2021 to Nov 2022

Publications

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. (2026). Application of machine learning techniques in the measurement of ground reaction force (GRF) waveforms using inertial measurement unit (IMU) sensors: a systematic literature review. *Medical Engineering & Physics*. doi
- Rizvi, D. Q., Majeed, M., **Manj, U. J.**, & Haq, M. I. U. (2025). Sleep disorder prediction with PSO-optimized machine and deep learning models. *ICODT2*, pp. 1 to 6. doi
- Akhter, M. N., **Manj, U. J.**, Ali, H., Niazi, I. K., & Taylor, D. Deep learning-based kinetics and center of pressure estimation during sit-to-walk transitions from a single sacrum-mounted IMU. *IEEE Transactions on Biomedical Engineering* (under review).
- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. Predicting ground reaction force during sit to walk: towards kinetically instrumenting the Timed Up and Go test. *Medical & Biological Engineering & Computing* (under review).
- **Manj, U. J.**, Rizvi, S. D. Q., Niazi, I. K., Akhter, M. N., & Ali, H. Gated feature fusion and age confounding in plantar thermography for diabetes screening. *Biomedical Physics & Engineering Express* (under review).

- **Manj, U. J.**, Qamar, D., Rashid, M. A., Perwaiz, N., & Niazi, I. K. Speaker leakage and subject level accuracy in voice biomarkers for Parkinson's disease. *Machine Learning: Health* (under review).

Education

PhD	**Auckland University of Technology**, Health & Artificial Intelligence, Auckland, New Zealand	Oct 2025 to Present
MS	**National University of Sciences & Technology (NUST)**, Computational Science & Engineering, Islamabad • Thesis: energy optimisation in data centres using fuzzy inference systems (~30% cost reduction).	2017 to 2021
BE	**National University of Sciences & Technology (NUST)**, Computer Engineering, Islamabad • Thesis: NATO narrowband waveform on a software defined radio (digital signal processing, convolutional encoding).	2012 to 2016

Skills

Neurophysiology & Signals: EEG and IMU signal analysis, vibrotactile stimulation protocols, spectral and time frequency analysis

Machine Learning & Statistics: PyTorch, scikit-learn, subject level evaluation, leakage and confounding checks, SHAP, G*Power

Programming Languages: Python, C++, Bash, PowerShell

Research Practice: Participant data governance (Privacy Act 2020, ISO/IEC 27001), systematic review, scientific writing

Referees

Dr. Nazia Perwaiz, MSCA Industrial Postdoctoral Fellow, School of Computer Science, University of Galway, Ireland; Assistant Professor, SEECS, NUST. nazia.perwaiz@seecs.edu.pk

Dr. Haider Ali, Faculty of Design and Creative Technologies, Auckland University of Technology; AI Engineer, KPMG New Zealand. haider.ali@autuni.ac.nz